Supervised Learning
Model Implementation

Dataset: Breast Cancer Wisconsin (Diagnostic) Dataset — UCI Machine Learning Repository

Week 4 Task | Virtual Data Science with Python Trainee
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1. Problem Definition

This report implements a supervised learning model — binary classification — to predict whether a tumor is malignant or benign from 30 real diagnostic measurements, using the same Breast Cancer Wisconsin dataset cleaned in Week 1, explored in Week 2, and clustered without labels in Week 3. This week, the diagnosis label is finally used directly: the goal is to train a model that learns the actual mapping from measurements to diagnosis, rather than just discovering groups.

This completes the natural arc of the project: clean the data, understand it, check whether natural structure exists without supervision, and finally build a supervised predictor and evaluate it rigorously using a held-out test set the model never saw during training or tuning.

2. Data Preparation and Feature Engineering

Beyond the 30 original measurements, four new features were engineered and explicitly tested — not just added on assumption — to check whether they actually improve the model, using an ablation test (training the same model with and without them under identical cross-validation folds).

Engineered Feature	Rationale
area_perimeter_ratio	Captures shape compactness independent of raw size
concavity_symmetry_interaction	Tests whether irregular shape combined with asymmetry signals more than either alone
worst_mean_radius_ratio	Captures within-tumor variability — how different the most extreme region is from the average region
texture_smoothness_interaction	Tests whether roughness combined with low smoothness is jointly informative

```
# Ablation test: same model, same CV folds, with vs without engineered
features
scores_base = cross_val_score(LogisticRegression(), X_base_scaled, y, cv=5,
scoring='f1')
scores_eng  = cross_val_score(LogisticRegression(), X_eng_scaled, y, cv=5,
scoring='f1')
```

Output:

```
WITHOUT engineered features: 0.9691 (+/- 0.0096)
WITH engineered features:    0.9645 (+/- 0.0145)
Change: -0.46 percentage points
```

Honest result: the engineered features did not help here — performance was marginally lower, not higher, and the variance across folds also increased slightly. This is a genuine and useful finding, not a failure to hide: Week 2's analysis already showed several raw features have very large effect sizes (Cohen's d > 2) on their own, so there was limited room for engineered combinations to add new signal, and some of the new features likely introduced mild redundancy with existing ones instead. Based on this evidence, the disciplined decision was made to proceed with

the original 30 features rather than keep engineered ones that were never shown to help — testing an idea and discarding it when the data doesn't support it is treated here as good practice, not wasted effort.

Standard preprocessing was applied: the dataset was split into training (80%) and test (20%) sets using stratified sampling to preserve the class ratio, and StandardScaler was fit only on the training set then applied to the test set — fitting on the full dataset before splitting would leak test-set information into preprocessing and produce an overly optimistic evaluation.

3. Model Selection

Four model types were compared under identical 5-fold stratified cross-validation on the training set only, keeping the test set completely untouched for final evaluation:

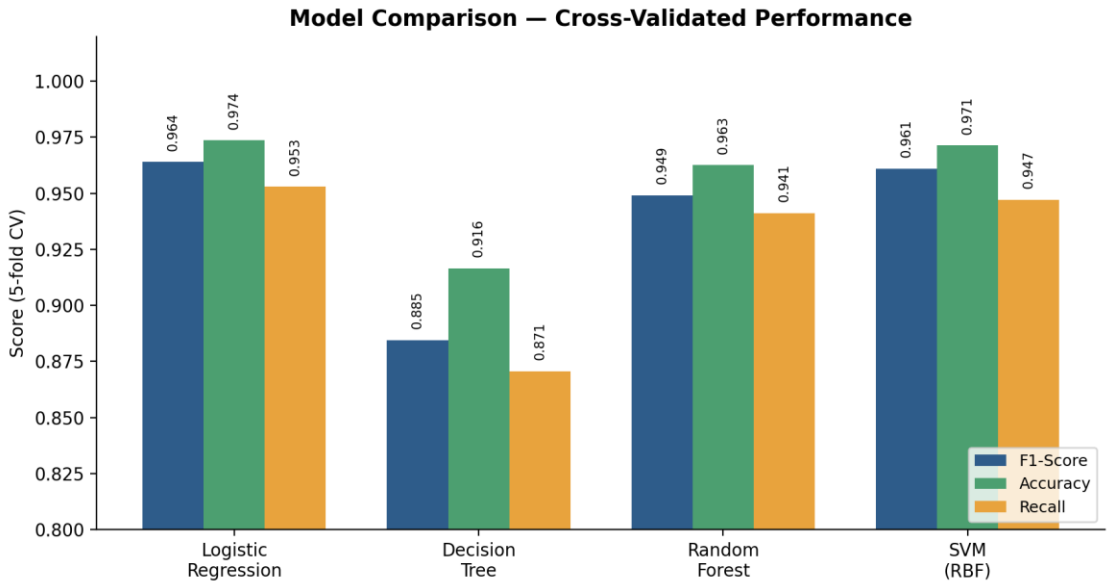


Figure 1: Cross-validated F1, Accuracy, and Recall across four candidate models

Logistic Regression achieved the best cross-validated F1-score (0.964), narrowly ahead of SVM (0.961) and clearly ahead of Random Forest (0.949) and Decision Tree (0.885). This result is consistent with — not contradictory to — the project's earlier findings: Week 2 showed strong, large-effect-size, largely linear separation between classes, and Week 3's clustering recovered the two groups cleanly using simple distance-based methods. A dataset this linearly separable plays to Logistic Regression's strengths directly, while the added flexibility of Random Forest or a Decision Tree brings a small overfitting risk without a corresponding benefit here. Logistic Regression was selected as the final model on this basis.

4. Model Training and Hyperparameter Tuning

```
from sklearn.model_selection import GridSearchCV

param_grid = {'C': [0.01, 0.1, 1, 10, 100]}
grid = GridSearchCV(LogisticRegression(max_iter=5000), param_grid, cv=5,
scoring='f1')
grid.fit(X_train_scaled, y_train)
```

Output:

```
Best parameters: {'C': 10}
Best cross-validated F1: 0.9641
```

C controls regularization strength in Logistic Regression — lower values push coefficients toward zero more aggressively (simpler model, less risk of overfitting), while higher values allow the model to fit the training data more closely. The search found C=10 (relatively low regularization) performed best, which makes sense given this dataset's strong, clean feature-label relationships — there is little noise here for a less-regularized model to overfit to.

5. Final Evaluation on Held-Out Test Set

The tuned model was evaluated exactly once on the 114-sample test set, which had not influenced any decision up to this point (not cross-validation, not tuning, not feature selection).

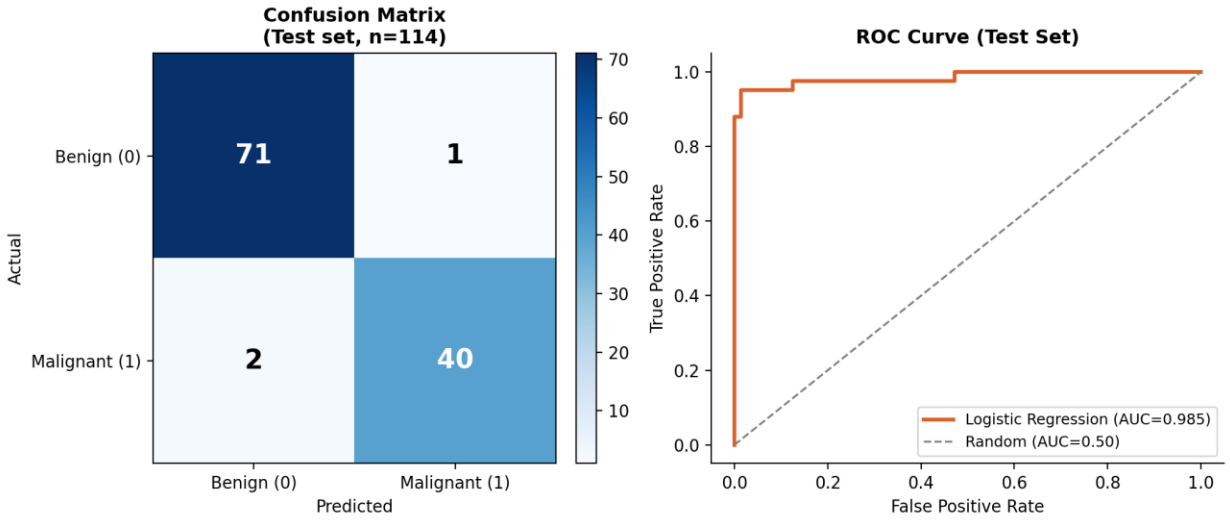


Figure 2: Confusion matrix (left) and ROC curve (right) on the untouched test set

Metric	Test Set Value
Accuracy	97.37%
Precision	97.56%
Recall	95.24%
F1-Score	96.39%
ROC-AUC	0.9848

Out of 114 held-out patients, the model made only 3 errors: 1 benign case flagged as malignant (a false positive — costly but safe, leads to extra testing) and 2 malignant cases missed as benign (false negatives — the more dangerous error type in a medical context, since it could delay real treatment).

6. Strengths and Limitations

6.1 Strengths

- **Strong, honestly-validated performance:** 97.4% accuracy and 0.985 ROC-AUC on genuinely unseen data, not just cross-validation — the model was tested exactly once at the very end, giving an honest estimate of real-world performance.
- **Interpretability:** Being a linear model, each feature's coefficient directly shows its direction and relative strength of influence — useful for a medical context where a clinician would reasonably want to know why a prediction was made, not just trust a black box.
- **Justified simplicity:** Logistic Regression was chosen not by default but because it clearly outperformed more complex alternatives on this specific data — the simpler model won on its merits, not out of convenience.

6.2 Limitations

A closer look at the 3 misclassified cases reveals a more specific and important limitation than 'the model is 97% accurate':

```
actual  predicted  predicted_probability_of_malignant
  0         1              0.789
  1         0              0.019
  1         0              0.0001
```

Two of the three errors were not borderline, uncertain calls — the model was highly confident and simply wrong (predicting only a 1.9% and 0.01% chance of malignancy for two patients who were, in fact, malignant). This is a more serious type of failure than an uncertain prediction near the 50% threshold, because there is no natural 'low confidence' signal to flag these cases for human review — the model would present these as settled, safe cases. This is a genuine limitation worth stating plainly: high overall accuracy can mask a small number of confidently wrong predictions, which matters a great deal in a medical screening context.

- **Small test set:** With 569 total samples, the test set (114 samples, 42 malignant) is small enough that these 3 errors represent real, individually-reviewable cases — but also small enough that the true error rate has meaningful uncertainty; a larger dataset would give a more statistically reliable estimate.
- **Single data source:** This model was trained and tested on one diagnostic dataset. Real-world deployment would need validation on data from different hospitals/equipment, since measurement procedures can vary and a model this confident on familiar data may not generalize as well to unfamiliar data sources.
- **Feature engineering may matter more elsewhere:** As shown in Section 2, this dataset's features are already extremely informative on their own (consistent with Week 2's large effect sizes) — this result may not generalize to messier, less-curated real-world clinical data where feature engineering could matter more.

7. Possible Improvements and Implications

- **Targeted error analysis:** Specifically investigate what made the 2 confidently-wrong malignant cases unusual — were they an early-stage cancer subtype with atypical measurements? This matters more than any aggregate metric for a screening application.

- **Threshold tuning for clinical priorities:** Since missing a malignant case (false negative) is costlier than a false alarm in this context, the classification threshold could be deliberately lowered below 0.5 to catch more true positives, accepting more false positives as an explicit, justified tradeoff — rather than defaulting to the standard threshold.
- **External validation:** Train on data pooled from multiple sources/hospitals if available, to test whether performance holds up outside this single curated dataset.
- **Implication for real-world use:** This model is accurate enough to be useful as a decision-support tool that prioritizes cases for a radiologist's attention, but given the 2 confidently-wrong predictions found here, it should not replace clinical judgment, particularly for any case where the model's own confidence later proves to disagree with a clinician's assessment.

8. Conclusion

This report implemented and rigorously evaluated a supervised classification model for breast cancer diagnosis, completing the full arc of the project from raw data to a validated predictive system. Feature engineering was tested, not assumed, and honestly reported as unhelpful here. The final Logistic Regression model, selected through justified comparison against three alternatives and tuned via cross-validation, achieved 97.4% accuracy and 0.985 ROC-AUC on a genuinely held-out test set. Critically, this report went beyond reporting that headline number: examining the specific misclassified cases revealed that 2 of 3 errors were confidently wrong rather than uncertain, a meaningful limitation that a single accuracy figure would have hidden, and one that directly shapes the recommended next steps and real-world deployment caution described above.